REHAN SHAIK

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PROFILE

Computer Science Engineering student (graduating 2028) at VIT-AP University with strong foundations in probability, statistics, linear algebra, and data structures. Experienced in building end-to-end ML pipelines, convolutional neural networks, and AI-powered systems. Seeking to deepen expertise in supervised learning, deep learning, and NLP through structured research and mentorship.

TECHNICAL SKILLS

ML / AI: Machine Learning, Deep Learning, CNN, Computer Vision, scikit-learn, TensorFlow/Keras, Supervised Learning, Classification, Model Evaluation, Data Augmentation, Machine Learning Pipelines, Gemini API

ML Foundations: Probability & Statistics, Linear Algebra

Programming: Python, Java, C

Core CS: Data Structures and Algorithms, Object-Oriented Programming (OOP), REST APIs, SQL

Web Technologies: JavaScript, TypeScript, Angular, Next.js, HTML, CSS, Bootstrap

Tools & Platforms: Git, GitHub, ESP32, IoT Development

EDUCATION

B.Tech – Computer Science Engineering | VIT-AP UNIVERSITY **|** CGPA: **8.87 |** 2024 – Present **|** Expected Graduation: 2028

Intermediate – NARAYANA JUNIOR COLLEGE **| 96% |** 2022 – 2024

Secondary School – BHASHYAM HIGH SCHOOL **| 95.6% |** 2021 – 2022

PROJECTS

Saurashtra Character Recognition System — OCR with Deep Learning | *Python · CNN · Computer Vision* **|** GitHub

- Built an end-to-end OCR pipeline for the low-resource Saurashtra script using CNN-based deep learning and computer vision.
- Constructed a custom labeled dataset of **850+ images across 85 character classes**; designed and trained a CNN classification **model achieving 99% training accuracy and 97–98% validation accuracy.**
- Applied data augmentation, dropout regularization, and batch normalization to improve generalization on unseen handwritten characters.
- Developed a real-time GUI application for character prediction with confidence-score outputs, making the model accessible to end users.

Genesis — AI-Powered Application Schema Generation Platform | *Next.js · TypeScript · Gemini API* **|** GitHub

- Designed a 4-stage Gemini API pipeline (**intent extraction → architecture generation → schema creation → validation**) that converts natural language prompts into validated application schemas with live previews.
- Engineered automated schema conflict detection and repair mechanisms, reducing invalid AI-generated outputs to near zero.
- Implemented a runtime renderer and analytics dashboard tracking real-time generation metrics and schema quality insights.
- Integrated REST APIs with structured prompt engineering to ensure consistent, well-formed outputs from an LLM.

Sentinel — Smart Safety Choker | *ESP32 · IoT · Embedded Systems · JavaScript* **|** GitHub

- Built an ESP32-based IoT safety wearable combining GPS tracking, heart-rate monitoring, and SOS alerting with real-time sensor data processing.
- Implemented stress detection and blackout-zone analysis using signal processing techniques on live sensor streams.
- Developed a responsive web dashboard for live GPS, heart-rate, and system-status visualization, with end-to-end communication between hardware, APIs, and frontend.

ACHIEVEMENTS & ACTIVITIES

- Pitched **TinkedOut**, an ML-powered engineer collaboration platform, at the **Wadhwani Entrepreneurship Portal, VIT-AP**.
- Active Member, Machine Learning Club, VIT-AP University

CERTIFICATIONS

- **Certified By Infosys Springboard as Angular Web Developer** – Apr 2026
- **Certified By Forage in Software Engineering Job Simulation** – Mar 2026

LANGUAGES

English (Professional) **|** Hindi **|** Telugu